

PROBLEM SET 2

→ Task 1

```

riscv64-unknown-elf-ld -z max-page-size=4096 -T user/user.ld -o user/_secret use
r/secret.o user/lib.o user/usys.o user/print.o user/umalloc.o
riscv64-unknown-elf-objdump -S user/_secret > user/secret.asm
riscv64-unknown-elf-objdump -t user/_secret | sed '1,/SYMBOL TABLE/d; s/ .* // ;
//s/d/ > user/secret.sym
mkfs/mkfs fs.img README user/exec.sh user/cat user/echo user/forktest user/g
rep user/init user/kill user/ls user/mkdir user/rm user/sh user/
stressfs user/userstests user/grind user/wc user/zombie user/logstress user/
forphan user/dorphan user/attack user/secret
mmapfs 47 (boot: super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1
953 total 2000
ballocc: first 988 blocks have been allocated
ballocc: write bitmap block at sector 46
*** Now run 'gdb' in another window.
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp
3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,f
ormat=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0 -S -gdb
tcp:20000

xv6 kernel is booting
hart 1 starting
hart 2 starting
[]

kernel/syscall.c
2 #include "types.h"
3 #include "user.h"
4 #include "kernel.h"
5 #include "syscalls.h"
6 #include "unistd.h"
7 #include "fcntl.h"
8 #include "stdio.h"
9
10 // Fetch the uint64 at addr from the current process.
11 int
12 fetchaddr(uint64 addr, uint64 ip)
13 {
14     struct proc *p = myproc();
15 }

remote Thread 1.3 (src) in: syscall
L133 PC: 0x80001c0b
(gdb) backtrace
#0 syscall () at kernel/syscall.c:133
#1 0x00000000 in usertrap () at kernel/trap.c:68
#2 0x00000000 in ?? ()
(gdb)

130
131 void
132 syscall(void)
133 {
134     int num;
135     struct proc *p = myproc();
136
137     num = p->trapframe->a7;
138     if (num < 0) num = 0;
139     // Use num to lookup the system call function for num, call it.
140     // and store its return value in p->trapframe->a0
141     p->trapframe->a0 = syscalls[num];
142     else
143         return;
144 }

remote Thread 1.3 (src) in: syscall
L137 PC: 0x80001c1f
#0 0x3fffffd080: 0 = 0x80007c00, 1 = 0x80007a00, 2 = 0x0,
s4 = 0x8012d00, s5 = 0x3, s6 = 0x8018970, s7 = 0x0, s8 = 0x8018a58,
s9 = 0x10, s10 = 0x0, s11 = 0x2, s12 = (0x0 repeats 16 times),
--Type <RET> for more, q to quit, c to continue without paging--
s16 = 0x8015d00, name = (0x0, 0x0, 0x0, 0x74, 0x0 <repeats 12 times>)
(gdb) p /X p->trapframe->a7
12 = 0x1
(gdb)

The kernel/syscall.c: warning: kernel/kernel has a LOAD segment with MMX perm
issions
riscv64-unknown-elf-objdump -S kernel/kernel > kernel/kernel.asm
riscv64-unknown-elf-objdump -t kernel/kernel | sed '1,/SYMBOL TABLE/d; s/ .* //
//s/d/ > kernel/kernel.sym
make[1]: Leaving directory /home/ubuntu-os/xv6-labs-2025/
== Test answers-syscall.txt ==
answers-syscall.txt: 0k
== Test sandbox_mask ==
$ make qemu-gdb
[]

Thread 3 hit Breakpoint 1, syscall () at kernel/syscall.c:133
133
(gdb) b *0x80001c1f
Breakpoint 2 at 0x80001c1f: file kernel/syscall.c, line 137.
(gdb) layout asm
Detaching from program: /home/ubuntu-os/xv6-labs-2025/kernel/kernel, process 1
Ending remote debugging.
[Inferior 1 (process 1) detached]
ubuntu-os@ubuntu-os-VirtualBox:~$ xv6-labs-2025$ phill -f qemu-system-riscv64
ubuntu-os@ubuntu-os-VirtualBox:~$ xv6-labs-2025$ nano answers-syscall.txt
ubuntu-os@ubuntu-os-VirtualBox:~$ xv6-labs-2025$

```

I forgot to take part 5 and part 6 picture but you can see answer in the answers-syscall text file and make grade also said that I did it correctly.

→ Task 2

```

ubuntu-os@ubuntuos-VirtualBox: ~/xv6-labs-2025
953 total 2000
ballocc: first 1027 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp
3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,f
ormat=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 2 starting
hart 1 starting
init: starting sh
$ sandbox 32768 README grep xv6 x
grep: cannot open x
$ sandbox 32768 - cat README
cat: cannot open README
$ ./grade-lab-syscall sandbox_mask
exec ./grade-lab-syscall failed
$ QEMU: Terminated
ubuntu-os@ubuntuos-VirtualBox:~/xv6-labs-2025$ ./grade-lab-syscall sandbox_mask
make: 'kernel/kernel' is up to date.
== Test sandbox_mask == sandbox_mask: OK (2.3s)
(Old xv6.out.sandbox_mask failure log removed)
ubuntu-os@ubuntuos-VirtualBox:~/xv6-labs-2025$

```

→ TASK 3

```

ubuntu-os@ubuntuos-VirtualBox: ~/xv6-labs-2025
953 total 2000
ballocc: first 1027 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp
3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,f
ormat=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ $ sandbox 32768 README grep xv6 README
# expected: contents of README related lines printed
exec $ failed
$ exec # failed
$ sandbox 32768 README grep xv6 README
xv6 is a re-implementation of Dennis Ritchie's and Ken Thompson's Unix
Version 6 (v6). xv6 loosely follows the structure and style of v6,
xv6 is inspired by John Lions's Commentary on UNIX 6th Edition (Peer
(kaashoek,rtm@mit.edu)). The main purpose of xv6 is as a teaching
$ sandbox 32768 README grep xv6 x
grep: cannot open x

```

→ TASK 4

```
ubuntu-os@ubuntuos-VirtualBox: ~/xv6-labs-2025
/^$/d' > user/attack.sym
mkfs/mkfs fs.img README user/exec.sh user/_cat user/_echo user/_forktest user/_g
rep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh user/_
stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_
_forphan user/_dorphan user/_sandbox user/_attack user/_secret
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1
953 total 2000
ballocc: first 1030 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp
3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,f
ormat=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 2 starting
hart 1 starting
init: starting sh
$ secret abc123
$ attack
N
$ attack
abc123
$
```

→ GRADE OF ALL TASKS

```
ubuntu-os@ubuntuos-VirtualBox: ~/xv6-labs-2025

== Test answers-syscall.txt ==
answers-syscall.txt: OK
== Test sandbox_mask ==
$ make qemu-gdb
sandbox_mask: OK (10.3s)
== Test sandbox_fork ==
$ make qemu-gdb
sandbox_fork: OK (1.3s)
== Test sandbox_path ==
$ make qemu-gdb
sandbox_path: OK (3.2s)
== Test sandbox_most ==
$ make qemu-gdb
sandbox_most: OK (1.4s)
== Test sandbox_minus ==
$ make qemu-gdb
sandbox_minus: OK (1.9s)
== Test attack ==
$ make qemu-gdb
attack: OK (1.5s)
      (Old xv6.out.attack failure log removed)
== Test time ==
```